
HYBRID GENERATIVE FRAMEWORK: INTEGRATING SINGLE-IMAGE GANS WITH CONTROLNET-GUIDED STABLE DIFFUSION FOR HIGH-FIDELITY IMAGE SYNTHESIS *

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ABSTRACT

Generative modeling has seen remarkable progress in recent years, driven mainly by two approaches: Generative Adversarial Networks (GANs) and Diffusion Models. GANs are known for capturing fine textures and structural details, while diffusion models excel at producing images with high fidelity and rich semantics. Yet, each has its limitations GANs often need large datasets and provide limited semantic control, while diffusion models require strong conditioning to preserve structural consistency. This study presents a hybrid framework that combines the strengths of both. A conditional single-image GAN is used to capture local textures and patterns from minimal data, while Stable Diffusion, guided by ControlNet, adds stylistic variety and realism. By conditioning the diffusion model with edge maps, the framework ensures structure is preserved while allowing flexible style transfer. Experiments show that this approach produces high-quality, controllable, and semantically rich images, even in data constrained scenarios. The results suggest that combining adversarial and diffusion-based methods offers a promising path toward efficient, versatile, and high-fidelity image generation.

Keywords GAN, Stable Diffusion, ControlNet, Canny Edge, Image Generation, Deep Learning

1 Introduction

GANs excel at learning from limited data but often produce artifacts and lack fine artistic control [1]. SinGAN demonstrated impressive single-image generation capabilities by learning internal patch distributions from just one image, enabling high-fidelity image manipulation and synthesis with limited data [11]. However, SinGAN and its variants often lack precise semantic control, limiting meaningful image editing [7]. Conversely, diffusion models like Stable Diffusion produce stunning visuals from text but require guidance for user-provided structure [2]. Recent advancements in diffusion models [4] have significantly bolstered the field of image generation [3] and have gained great success as generative models—especially in text-to-image generation tasks [5]. In the field of artificial intelligence, image-generation techniques have been a hotspot for research [50].

This study proposes a hybrid two-stage pipeline leveraging both GANs and diffusion models to produce images combining content fidelity and stylized expressiveness. First, a SinGAN-inspired conditional single-image

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GAN—which incorporates latent noise and additional feature vectors as conditioning inputs—learns and faithfully reproduces the low-level textures and structural details of an input photograph [8]. Then, a pretrained Stable Diffusion model, guided by a ControlNet edge-map adapter derived from Canny edge detection and driven by a user-supplied text prompt, transforms that structure into a richly detailed, artistically stylized output [9, 10]. This enables users to preserve precise form while rendering it with cinematic lighting, painterly brushstrokes, or other desired effects.

This work introduces a conditional single-image GAN that incorporates both latent noise and feature vectors to generate diverse yet structurally faithful outputs, a ControlNet module guided by Canny edge detection to provide precise structural conditioning for image synthesis, and an integrated pipeline that combines these components with a text prompt to produce final images that maintain the original form while enabling expressive, high-quality stylization.

Combining GANs and diffusion models leverages the structural learning of GANs with the prompt-driven creativity of diffusion models. This hybrid architecture is well-suited for flexible and artistic animation generation.

2 Related Work

The field of generative modeling has been dominated by two paradigms: Generative Adversarial Networks (GANs) and Diffusion Models, each offering unique strengths and limitations.

The foundation of adversarial learning was established by Goodfellow et al. [13], who introduced GANs as a framework where a generator and discriminator compete in a minimax game. GANs have since been widely applied in image generation, style transfer, and representation learning due to their ability to model high-dimensional data distributions. However, conventional GANs often require large datasets and suffer from instability in training.

To address data efficiency, Shaham et al. [11] proposed SinGAN, a model capable of learning internal patch distributions from a single natural image. This work demonstrated that GANs can produce diverse and realistic outputs without relying on large datasets, enabling creative applications such as image manipulation and texture transfer. Despite these advantages, SinGAN lacks semantic control, limiting its ability to generate images aligned with user intent.

In contrast, Diffusion Models emerged as a powerful alternative for generative tasks. Ho et al. [4] introduced Denoising Diffusion Probabilistic Models (DDPMs), which gradually denoise random noise to synthesize highly realistic images. Diffusion-based methods have since outperformed GANs in terms of fidelity and diversity, but they typically require strong conditioning mechanisms to preserve structural features when transforming input images.

A major advancement in controllable diffusion generation came from Zhang et al. [2], who introduced ControlNet as an extension to Stable Diffusion. By conditioning the generation process on structural information such as edge maps, ControlNet enables fine-grained control over the output while maintaining high visual quality. This method is particularly effective for tasks requiring structural consistency alongside stylistic variation.

Building on these foundations, this study introduces a hybrid framework that integrates a conditional single-image GAN with ControlNet-guided Stable Diffusion. The GAN component captures low-level textures and structural details from minimal data, while the diffusion model ensures semantic richness and stylistic expressiveness. This approach leverages the complementary strengths of both paradigms, achieving controllable and high-quality image synthesis even under limited-data scenarios.

3 Methodology

The experimental framework as illustrated in Figure 1, demonstrates a two-phase workflow combining a Generative Adversarial Network (GAN) and Stable Diffusion with ControlNet to generate a high-quality, stylized image. The process begins with a latent vector—random noise input—which is passed into the generator model. The generator, conditioned with feature and content information, produces a fake generated image that mimics the original input. Simultaneously, a real image is also provided. Both real and fake images are fed into the discriminator, which acts as a binary classifier to determine whether an image is "real" or "fake." Based on this, the discriminator model updates itself to better distinguish images, while the generator model updates to fool the discriminator, improving its outputs. This adversarial training loop continues until the final epoch. At this final stage, the generator's output is passed into a ControlNet + Stable Diffusion model, along with a Canny edge map and a text prompt, which together guide the final image generation. This hybrid architecture combines the content preservation strengths of GANs with the stylistic richness of diffusion models, resulting in a visually detailed and controllable output image.

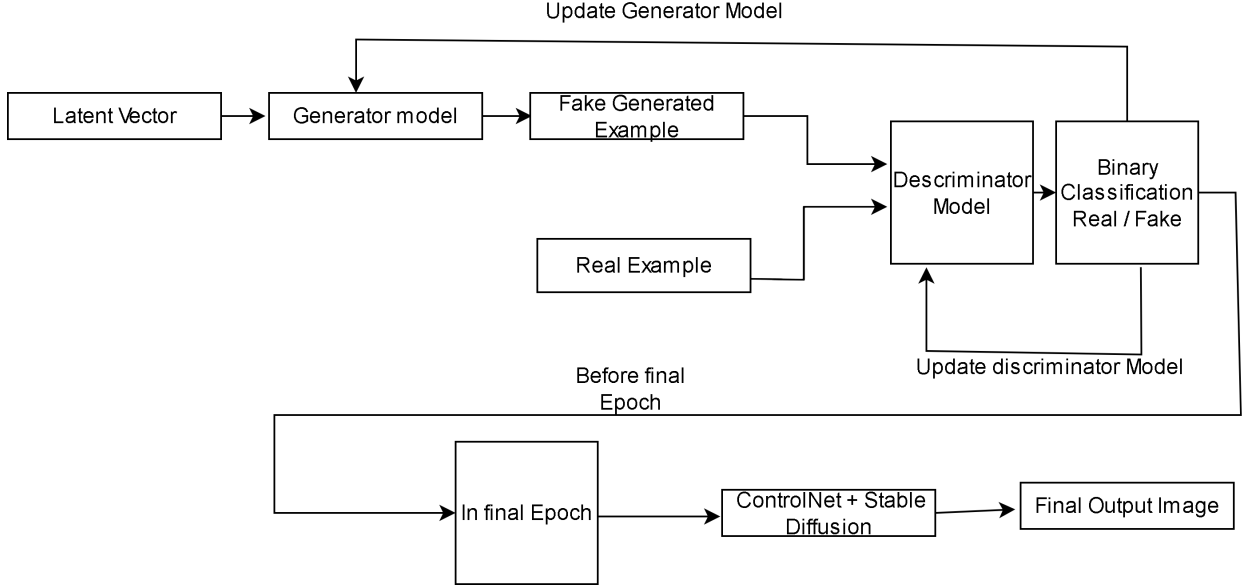


Figure 1: Experimental Framework combining GAN with Stable Diffusion and ControlNet. The architecture illustrates the generation of a stylized output image using a hybrid adversarial-diffusion pipeline guided by edge features and textual prompts.

4 Dataset Collection and Description

This work does not rely on large-scale datasets, instead, a single scratch image was used as input image to simulate a minimal-data scenario for creative content generation, aligning with approaches based on single-image generative modeling. [11, 12] The approach is intended to support animators and digital artists, enabling them to explore and synthesize visual ideas from a single reference image. By integrating generative adversarial networks (GANs) and diffusion models, the framework facilitates the generation of diverse visual outputs that can serve as inspiration for character development, scene concepts, or artistic direction.

4.1 Image Preprocessing

Image preprocessing is a critical step in ensuring that both the GAN and diffusion components of the framework receive appropriately formatted input, which directly impacts the quality and stability of generated outputs [13, 14]. The original RGB image is processed in two distinct ways to suit the specific requirements of the GAN and ControlNet-guided Stable Diffusion pipelines.

GAN Input Preprocessing

For the GAN component, the image undergoes the following transformations:

- **Color Mode Conversion:** The image is first loaded and explicitly converted to RGB to ensure consistent color channels, as recommended in standard image processing libraries such as Pillow [15].
- **Resizing:** The image is resized to a spatial resolution of 128×128 pixels, which matches the expected input size for the custom generator and discriminator networks. This smaller size reduces computational load and enables faster training [16, 17].
- **Normalization:** The pixel values are normalized to the range $[-1, 1]$ [20], using the transformation:

$$x_{\text{norm}} = \frac{x}{0.5} - 1 \quad (1)$$

This normalization aligns with the generator’s use of the Tanh activation function, which outputs values in the same range. It ensures stable learning dynamics and helps accelerate convergence during GAN training [18, 19].



Figure 2: The scratch image used as input for training and generation.

ControlNet Input Preprocessing

The same input image is also prepared for use as a conditioning input to the ControlNet model in the Stable Diffusion pipeline. The steps are as follows:

- **Resizing:** The image is resized to 512×512 pixels, the standard input resolution required by the ControlNet and Stable Diffusion models. This larger size helps preserve finer structural details, [21, 2] supported by recent findings on resolution-adaptive conditioning mechanisms [22].
- **Canny Edge Detection:** Using OpenCV’s `cv2.Canny()` function with thresholds of 100 and 200, edges are extracted from the image. This highlights significant object boundaries and intensity transitions [23]. Canny edges play a crucial role as a conditional input for the ControlNet model [24].
- **Edge Map Conversion:** The resulting edge map is a single-channel grayscale image. To match ControlNet’s expected 3-channel RGB input, the grayscale image is converted into a 3-channel format by duplicating the edge values across all channels. This preserves structural information while ensuring compatibility [25].

The dual preprocessing approach ensures that the GAN learns structure-aware transformations on a compact 128×128 input, while ControlNet receives a high-resolution (512×512) edge map that guides Stable Diffusion to produce detailed and creative images that preserve the original image’s structural layout [26].

4.2 GAN Architecture

Generative Adversarial Networks GANs [13, 31] have shown impressive performance across various domains, particularly in image generation [32, 33], image editing [34], and representation learning [33, 35, 36]. The proposed Generative Adversarial Network (GAN) consists of two key components: the **Generator** and the **Discriminator** [13], each playing an adversarial role to improve image generation quality through iterative training.

Generator

The Generator is designed to produce realistic synthetic images from a combination of inputs:

- A 3-channel image (usually RGB),
- A latent vector of size 128 (random noise sampled from a normal distribution), and

- A feature vector of size 10 (representing specific content-related conditions such as emotion, style, or domain context).

The 128-dimensional latent vector enables the generator to produce diverse outputs under identical input conditions [37]. Both the latent vector and the feature vector are independently processed through fully connected (FC) layers [27]. These layers transform each vector into a spatial map of size 128×128 , aligning them with the spatial dimensions of the input image. The resulting two feature maps are then concatenated along with the 3-channel input image, forming a 5-channel composite tensor (3 image channels + 2 generated maps).

This 5-channel input is passed through a stack of six convolutional layers, each using a 3×3 kernel and ReLU activation functions [35]. These layers enable the model to capture both low-level and high-level features [30]. The final layer of the Generator applies a Tanh activation function, which maps the output pixel values to the range $[-1, 1]$, producing the final 3-channel synthetic image that visually resembles real input data. The Tanh function normalizes output pixels, helping generated images appear realistic within a set range [28].

Discriminator

The Discriminator serves as a binary classifier whose goal is to distinguish between real images and those generated by the Generator and in doing so, provides feedback that helps the Generator improve its output quality [38]. It accepts a 3-channel image as input—either a real image from the dataset or a fake image produced by the Generator.

The input image is processed through three convolutional layers, each with a 3×3 kernel and a LeakyReLU activation function. LeakyReLU is selected to prevent the ‘dying ReLU’ problem—a common challenge in GAN training—while ensuring gradient flow even for negative activations [39]. These layers progressively extract discriminative features that help identify the authenticity of the image.

Finally, the Discriminator applies a single convolutional layer with a Sigmoid activation function, which outputs a 1-channel probability map. This map represents the likelihood that different regions of the input image are real or fake. [40] The Discriminator is trained to output values close to 1 for real images and close to 0 for fake images.

4.3 Training Procedure

The training of the Generative Adversarial Network (GAN) involves an adversarial learning process between the Generator and the Discriminator [41]. The objective is for the Generator to produce increasingly realistic synthetic images while the Discriminator becomes better at distinguishing real from fake images [42, 43, 55].

Loss Function

The Binary Cross Entropy (BCE) [44] loss is used to train both components. For the Discriminator, this loss penalizes incorrect classification of real images as fake and vice versa. For the Generator, the loss encourages it to generate images that are classified as real by the Discriminator.

Optimizers

The Adam optimization algorithm is used for both the Generator and the Discriminator, the most popular adaptive method for GAN training [45]. The Adam optimizer parameters are set as follows:

- Learning rate $\alpha = 2 \times 10^{-4}$
- $\beta_1 = 0.5$
- $\beta_2 = 0.999$

Learning Rate Scheduling

To promote better convergence and prevent overfitting in later stages of training, the learning rate is reduced by half every 3,000 epochs.

Training Epochs

The model is trained for 10,000 epochs. In each epoch, the following steps are carried out:

- **Sampling:** A random latent noise vector (of size 128) and a binary feature vector (of size 10) are sampled. These serve as input to the Generator for synthesizing a new image.
- **Discriminator Update:** The Discriminator is trained using both real images (labeled as 1) and synthetic images produced by the Generator (labeled as 0). This step helps the Discriminator learn to distinguish real from generated images.

- **Generator Update:** The Generator is updated with the goal of “fooling” the Discriminator. It attempts to generate images that the Discriminator classifies as real (target label = 1), thereby improving its ability to generate high-quality, realistic outputs over time.

This iterative adversarial process allows both the Generator and the Discriminator to improve simultaneously, leading to more realistic image synthesis by the Generator [46].

4.4 Stable Diffusion with ControlNet

To further enhance the visual quality and stylistic consistency of the images generated by the GAN, this study employ a post-processing refinement stage using **Stable Diffusion** augmented with **ControlNet**. This combination enables the generation of high-resolution, photorealistic images guided by both structural constraints (in the form of edge maps) and semantic prompts (in natural language). GANs can generate highly realistic images but they often suffer from unstable training and mode collapse [48, 49], while diffusion models—especially Stable Diffusion—offer can improve image quality and training stability by leveraging a diffusion process along with conditional guidance in latent space [50].

Freezing the GAN: Once the Generative Adversarial Network (GAN) has completed training after the final epoch (e.g., at 10,000 epochs), all its weights are frozen. This ensures that its outputs remain stable and deterministic, and allows us to use the GAN-generated images solely as structural inputs for the next stage of synthesis.

ControlNet + Stable Diffusion Integration: In this refinement phase, we utilize two pre-trained models:

- **ControlNet:** A neural network extension that guides generative diffusion models using conditional inputs such as edge maps, depth maps, or pose annotations. Specifically, we use the `lllyasviel/control_v11p_sd15_canny` model, which is fine-tuned to condition generation on Canny edge maps.
- **Stable Diffusion v1.5:** A powerful text-to-image latent diffusion model provided by `runwayml/stable-diffusion-v1-5`, known for producing high-quality and coherent images from natural language descriptions [51, 52].

Input Preparation: The input to this diffusion pipeline includes:

- **Canny edge map:** Extracted from the GAN-generated image. This edge map provides structural guidance, preserving object contours and layout.
- **Text prompt:** This defines the desired semantic content and artistic style of the final image, e.g., “A water lady with ocean blue eyes, soft lighting, photorealistic portrait.”
- **Negative prompt:** A soft constraint to discourage unwanted artifacts (e.g., “blurry, deformed, extra limbs, low resolution”) [53].

Inference Parameters: The image generation process is conducted over 80 inference steps to ensure sufficient diffusion and detail refinement. A guidance scale of 12 is used to balance the influence of the text prompt against the conditioning edge map. Higher guidance values lead to images more closely aligned with the text description [54].

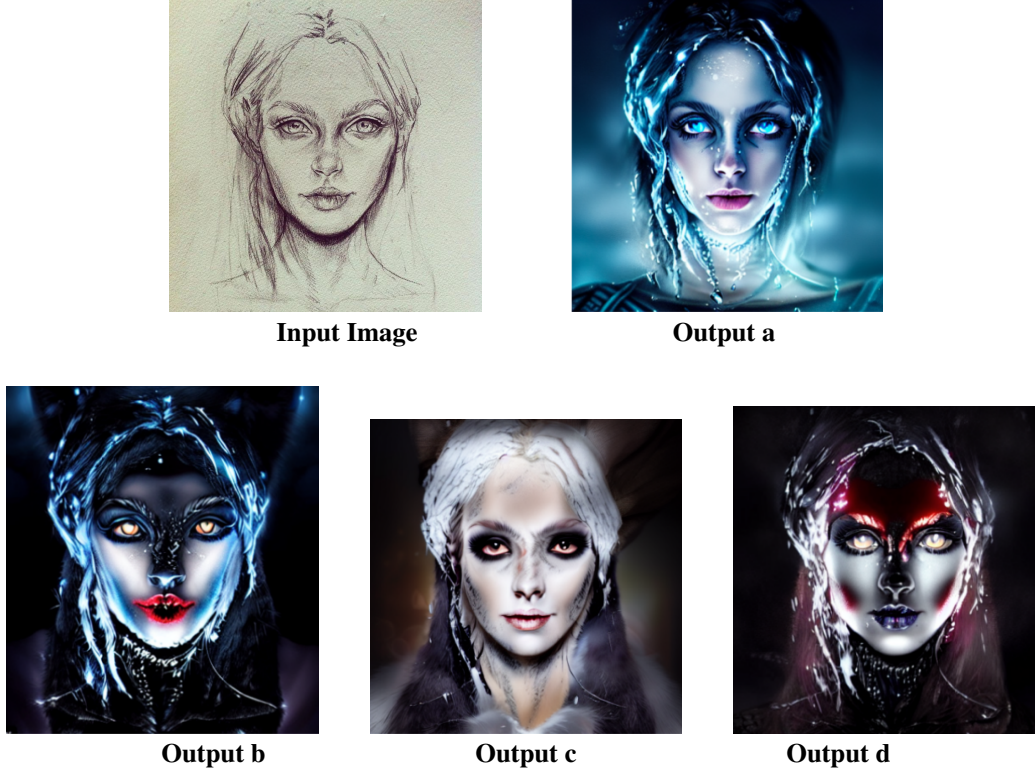
Output Rendering: After completion of the diffusion process, the final image is rendered and displayed. This image retains the structural fidelity of the original GAN output (due to the edge guidance), while significantly improving the visual quality, coherence, and stylization in accordance with the textual prompt.

Significance: This combined pipeline leverages the strengths of both GANs (fast, structure-aware generation) and diffusion models (fine-grained control and photorealism). By conditioning Stable Diffusion with ControlNet on the GAN’s output, this study achieve enhanced controllability and high-quality synthesis, making this an effective two-stage image generation strategy.

5 Experimental Results

5.1 Qualitative Examples

side-by-side comparisons of the original input images and the final outputs generated by the proposed hybrid pipeline, which combines a GAN and a Stable Diffusion model guided by ControlNet. Each output is conditioned on the corresponding edge map and a descriptive natural language prompt.



Prompt a: A water lady with ocean blue eyes, soft lighting, photorealistic portrait.

Prompt b: A supernatural hybrid entity, a fusion of a woman and a wolf with human and wolf-like traits, fur hinting, glowing eyes.

Prompt c: A supernatural hybrid entity, a fusion of a woman and a white fox with pointed ears, glowing slit-pupil eyes, soft fur.

Prompt d: A mythical red fox spirit woman with burning eyes, ethereal glow, and flowing fur cloak.

Figure 3: Comparison of one input image with four stylized outputs, each generated from a distinct prompt.



Input Image

Prompt: "A surreal fantasy scary fairy with horns, glowing eyes."



Final Output

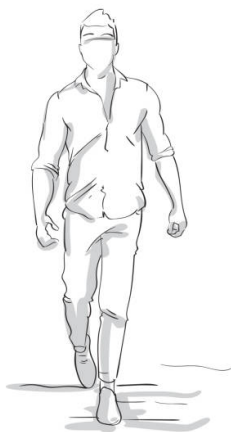


Input Image

Prompt: "A surreal fantasy mermaid with glowing eyes."



Final Output



Input Image

Prompt: "A man with joker face."



Final Output



Input Image

Prompt: “A fierce monkey man, confused face with red eyes, muscular build, and agile posture, blending human intelligence with primal wildness.”



Final Output



Input Image

Prompt: “a majestic golden cat with phoenix-like features standing above watter, glowing fiery two wings.”



Final Output



Input Image

Prompt: “a majestic moon fairy with phoenix-like features , glowing fiery two wings.”



Final Output

Figure 5: Visual comparison between the input and generated image for the prompt: “A man with monkey face.”

6 Conclusion

This work introduced a hybrid generative pipeline that combines single-image learning using Generative Adversarial Networks (GANs) with diffusion-based stylization guided by ControlNet. This approach enables high-quality image synthesis from limited data while leveraging the expressive power of prompt-based diffusion models for stylistic enhancement. This method demonstrates the potential of integrating GANs with controllable diffusion models to produce visually rich and diverse outputs. Future directions include conducting quantitative evaluations, extending the framework to multi-image or large-scale datasets, and incorporating interactive prompt editing to enhance user control over the generation process.

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